Exp 6: Automating Business Processes Using Deluge Scripting and Workflows, with Application Testing and Debugging

Register Number: 241401047

1. Aim

To automate the loan pre-screening process of the "Loan Application & Pre-Screening Management" application in Zoho Creator using Deluge scripting and workflows: enforcing a business rule at submission time, sending automated emails to applicants, and giving loan officers a one-click reminder action from the report. The scripts are then tested and debugged until they behave correctly.

2. Procedure

2.1 Automation Plan

Three workflows were built on the Loan_Application_Form, each using a different trigger type:

Workflow	Trigger	Purpose
Business_Hours_Callback_Validation	Record Created → Validations on form submission (runs before the record is saved and can cancel it)	Rejects an application whose Preferred_Callback_Time falls outside business hours (accepted window: 09:00 to 17:59).
Applicant_Acknowledgement_Email	Record Created → Successful form submission (runs after the record is saved)	Emails the applicant an acknowledgement using the Email_Address value.
Send_Document_Reminder_Workflow	On click of the "Send Document Reminder" action item in Loan_Application_Form Report	Lets a loan officer email a document-verification reminder to the applicant of the selected record.

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2.2 Open the Workflow Module

In Builder mode the Workflow tab was opened and the Form workflows section was used to create the first workflow.

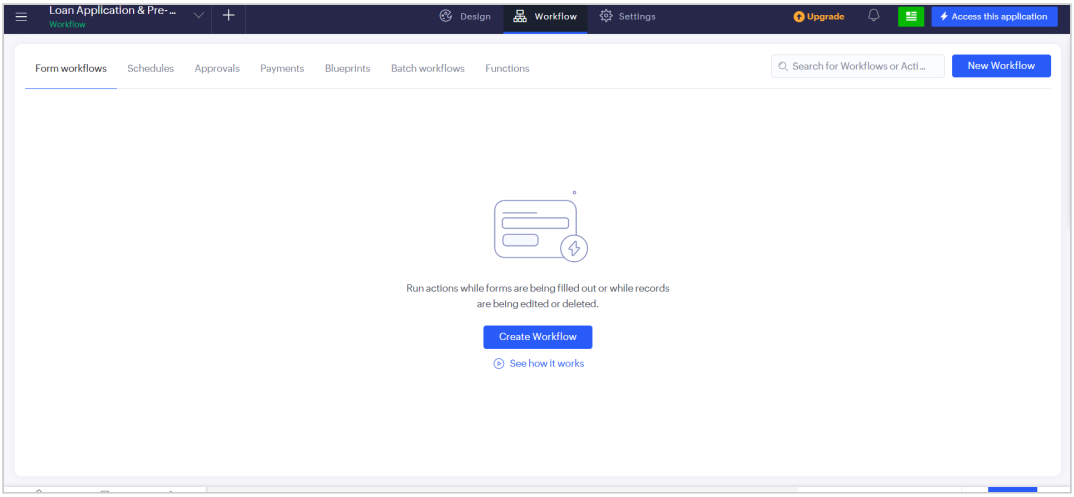
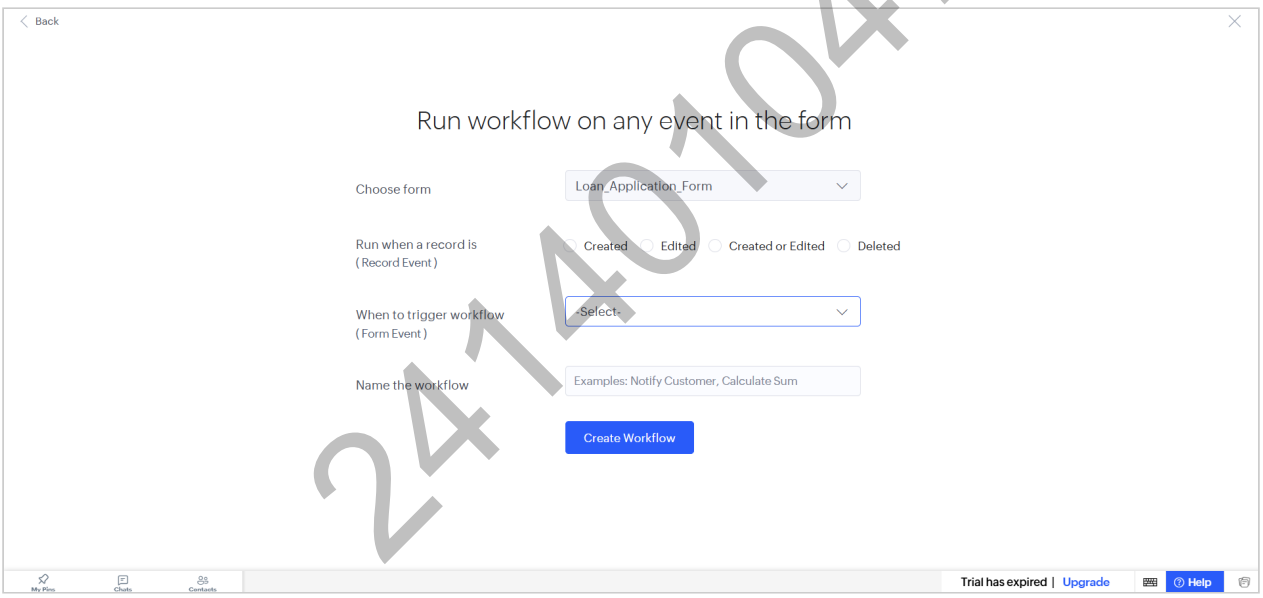


Figure 1: Workflow tab — Form workflows section with the Create Workflow button

2.3 Create the Validation Workflow

Loan_Application_Form was chosen as the form and "Created" as the record event, so the workflow runs when a new loan application is submitted.



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Figure 2: Create workflow screen — form and record event selection

The Form Event dropdown groups triggers into Before form submission, On form submission, and After form submission. "Validations on form submission" was selected for the business-hours rule because it can run a Deluge script and cancel the submission; "Successful form submission" is used later for the email.

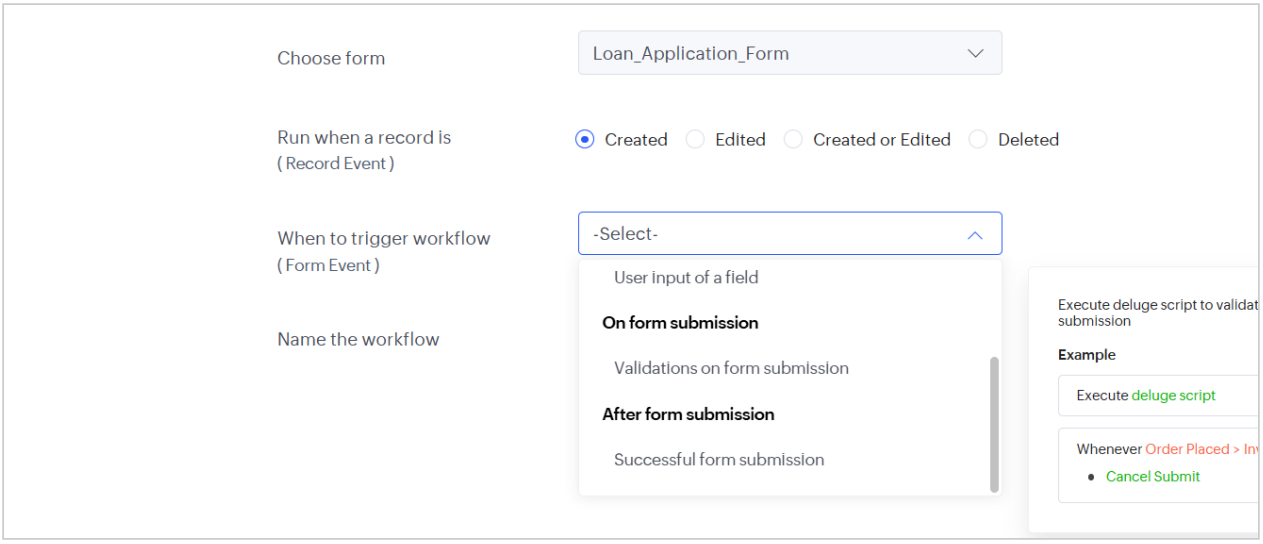


Figure 3: Form Event options — Validations on form submission and Successful form submission

2.4 Script 1 — Business Hours Validation

After naming the workflow Business_Hours_Callback_Validation, the Deluge script editor opened with a blank code area.

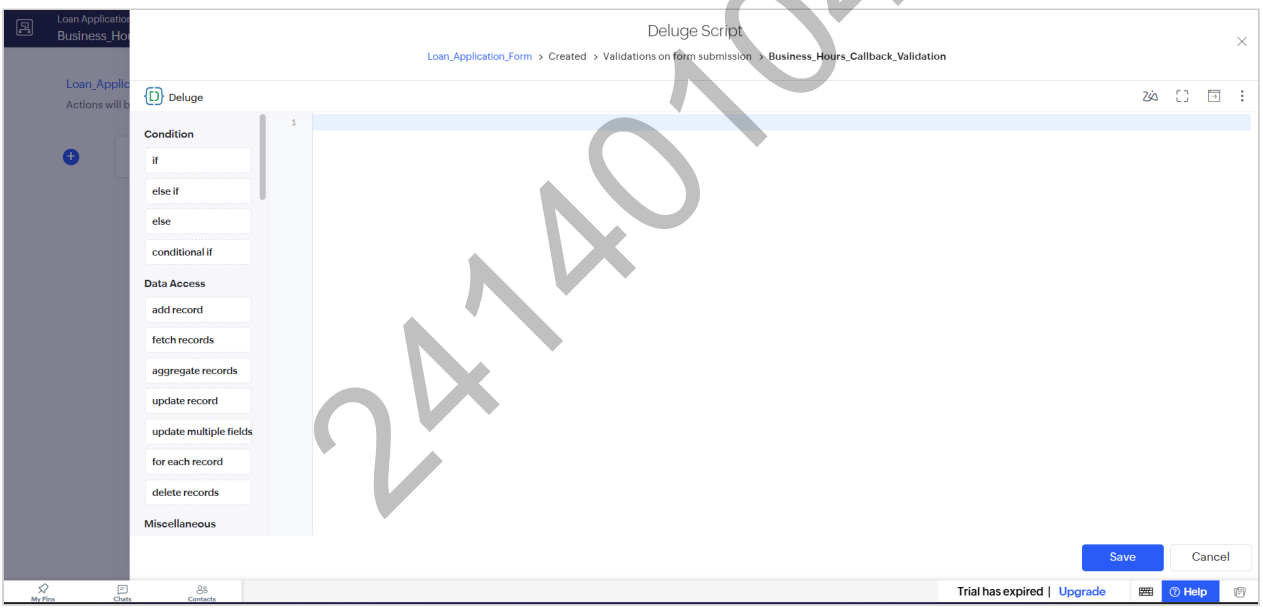


Figure 4: Deluge script editor for Business_Hours_Callback_Validation

The first version of the script converted the time value with toTime("HH") and toLong(). It saved without a syntax error, but failed when a record was submitted (see Section 3.2).

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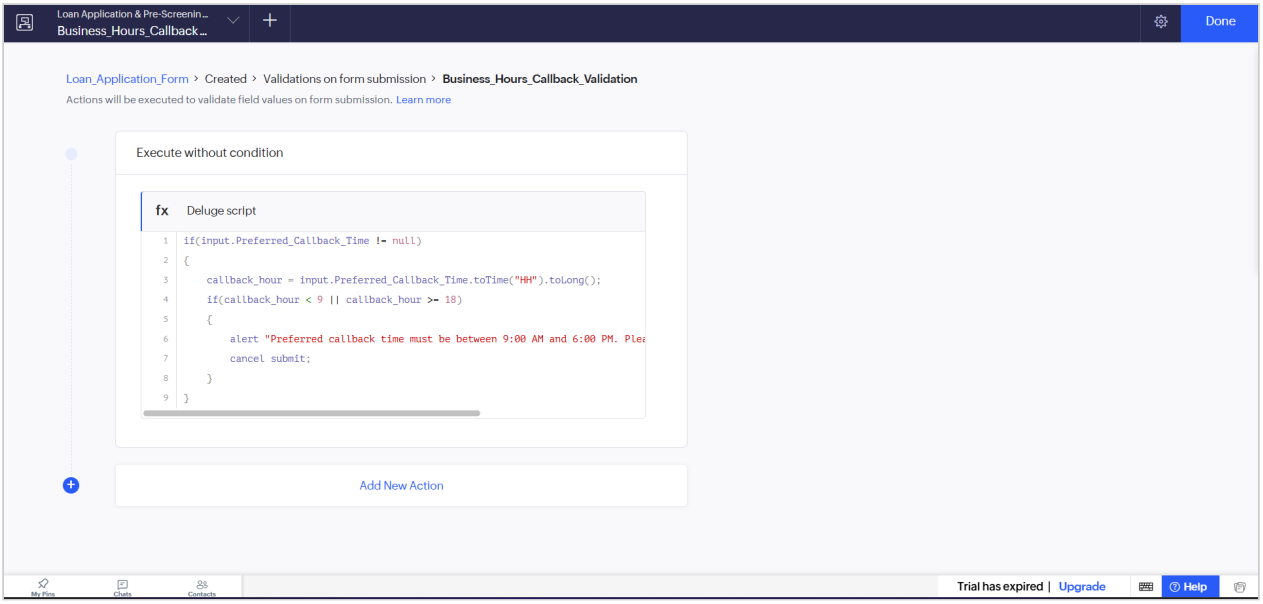


Figure 5: Script 1, first version saved in the workflow (before debugging)

Testing with this version produced the following run-time errors in the On Validate script:

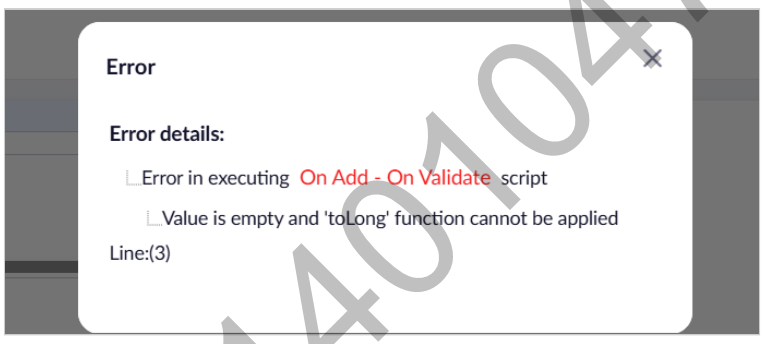


Figure 6: Error at line 3 — 'toLong' cannot be applied to an empty value

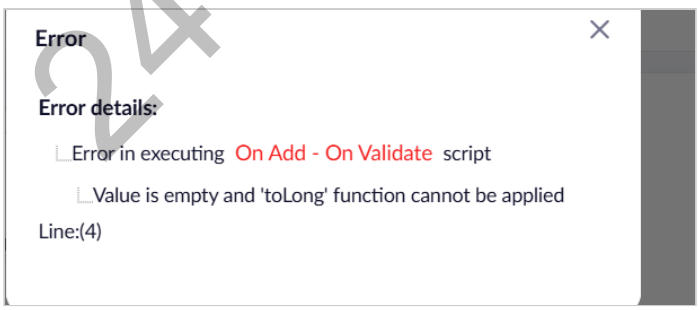


Figure 7: The same error at line 4 after changing the conversion to toString("HH")

The working version reads the first two characters of the time value (the hour), converts them to a number, and cancels the submission with an alert if the hour is before 9 or from 18 onwards:

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```
if(input.Preferred_Callback_Time != null)
{
    time_text = input.Preferred_Callback_Time;
    hour_text = time_text.subString(0,2);
    callback_hour = hour_text.toLong();
    if(callback_hour < 9 || callback_hour >= 18)
    {
        alert "Preferred callback time must be between 9:00 AM and 6:00 PM. Please
choose a time within business hours.";
        cancel submit;
    }
}
```

Script 1 (final version)

2.5 Script 2 — Applicant Acknowledgement Email

A second workflow, Applicant_Acknowledgement_Email, was created with the "Successful form submission" trigger, so the email is sent only after validation passes and the record is saved. In the Add New Action panel, "Always" was kept and the Deluge script action type was chosen.

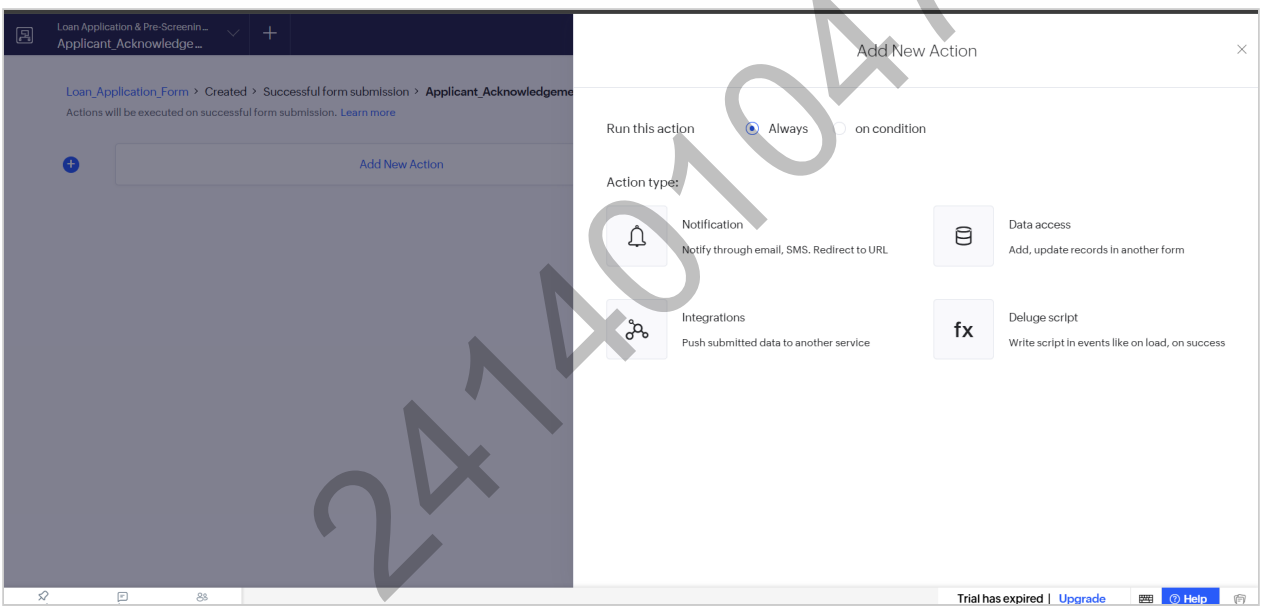


Figure 8: Add New Action panel — action types Notification, Data access, Integrations, and Deluge script

The first version used getPrefix() and getSuffix() on the Applicant_Name field and was rejected by the editor when saving:

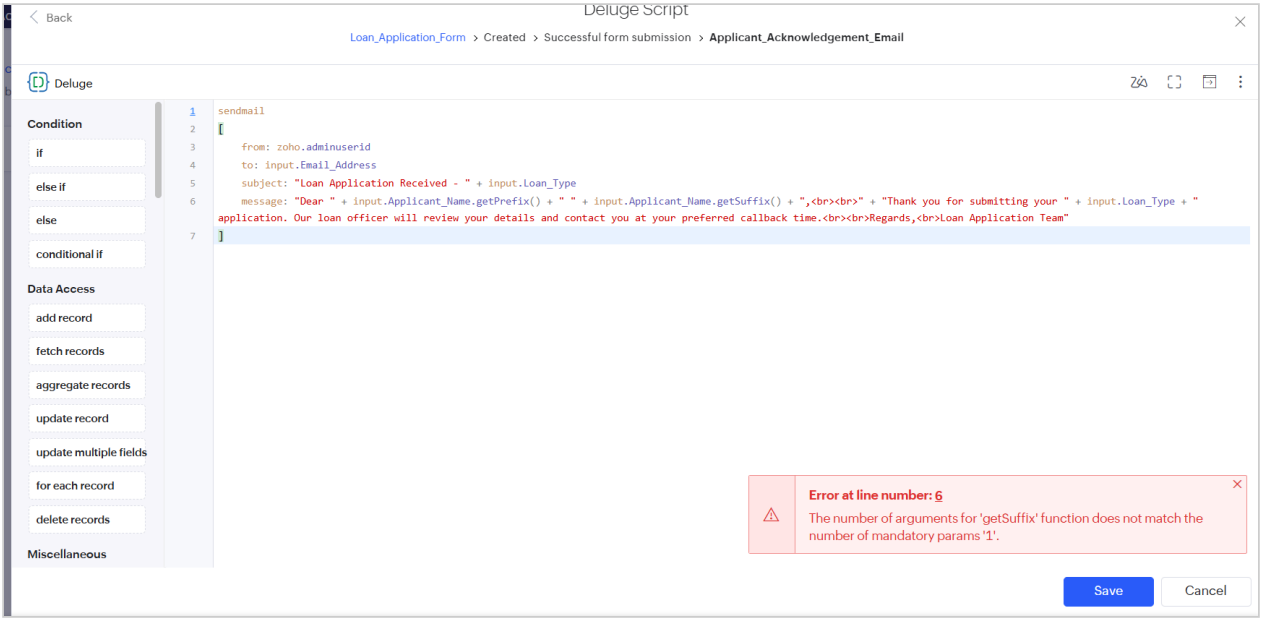
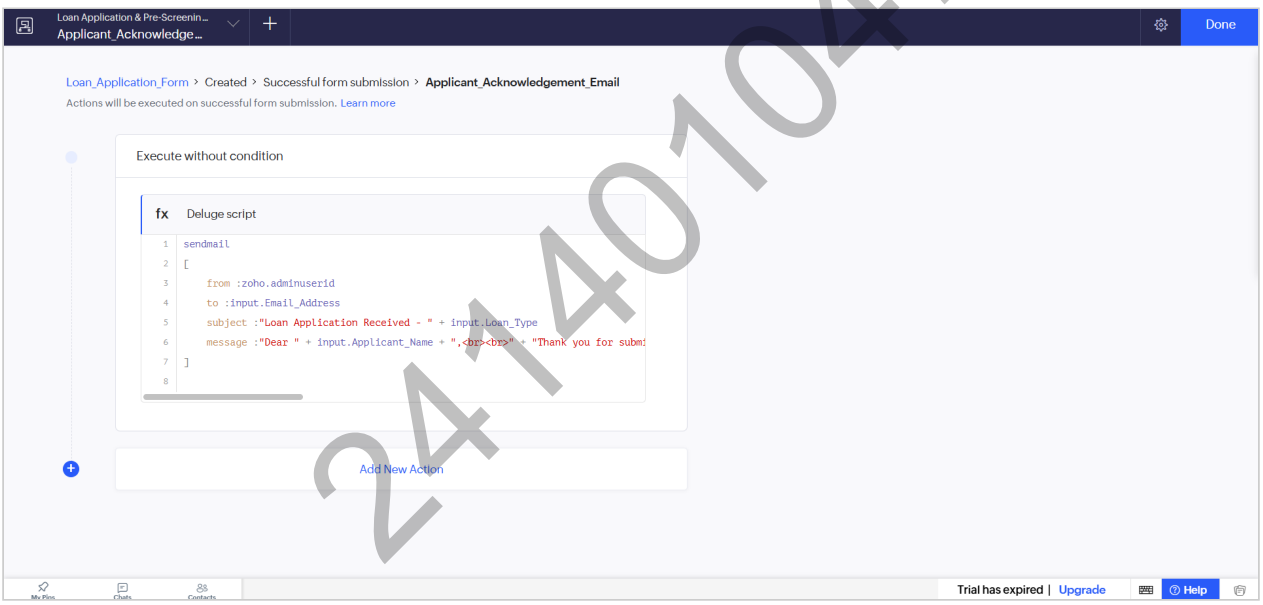


Figure 9: Editor error at line 6 — argument count mismatch for getSuffix

The corrected script uses the Name field directly and saved without errors:



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Figure 10: Script 2 (final version) saved in Applicant_Acknowledgement_Email

```
sendmail
[
    from: zoho.adminuserid
    to: input.Email_Address
    subject: "Loan Application Received - " + input.Loan_Type
    message: "Dear " + input.Applicant_Name + ",<br><br>" + "Thank you for submitting
your " + input.Loan_Type + " application. Our loan officer will review your details
and contact you at your preferred callback time.<br><br>Regards,<br>Loan Application
Team"
]
```

Script 2 (final version)

2.6 Script 3 — Send Document Reminder Action

In Design mode, Loan_Application_Form Report was opened and Report Customization → Actions → For Single Record was used to add a new action item named "Send Document Reminder". It is enabled Always and executes At once, and its workflow is Send_Document_Reminder_Workflow.

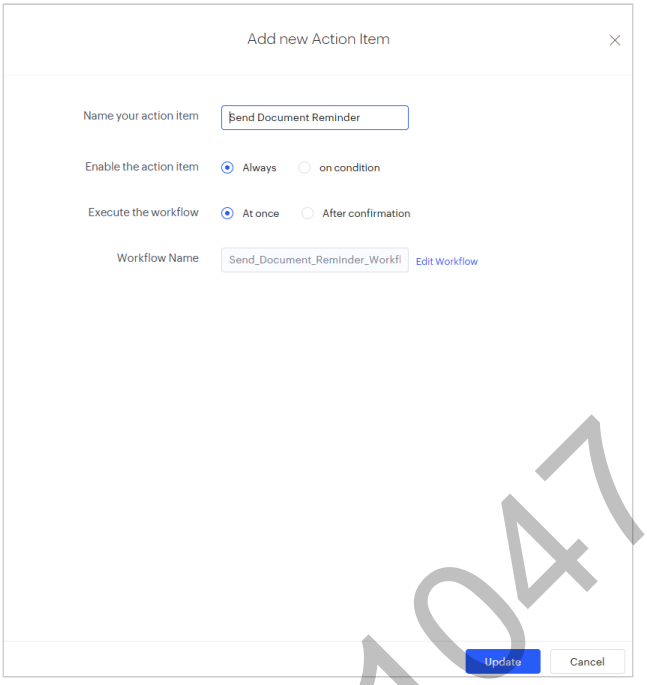


Figure 11: Action item configuration for "Send Document Reminder"

The workflow's script emails the applicant of the selected record. It is wrapped in a check so that no email is attempted when the record has no email address:

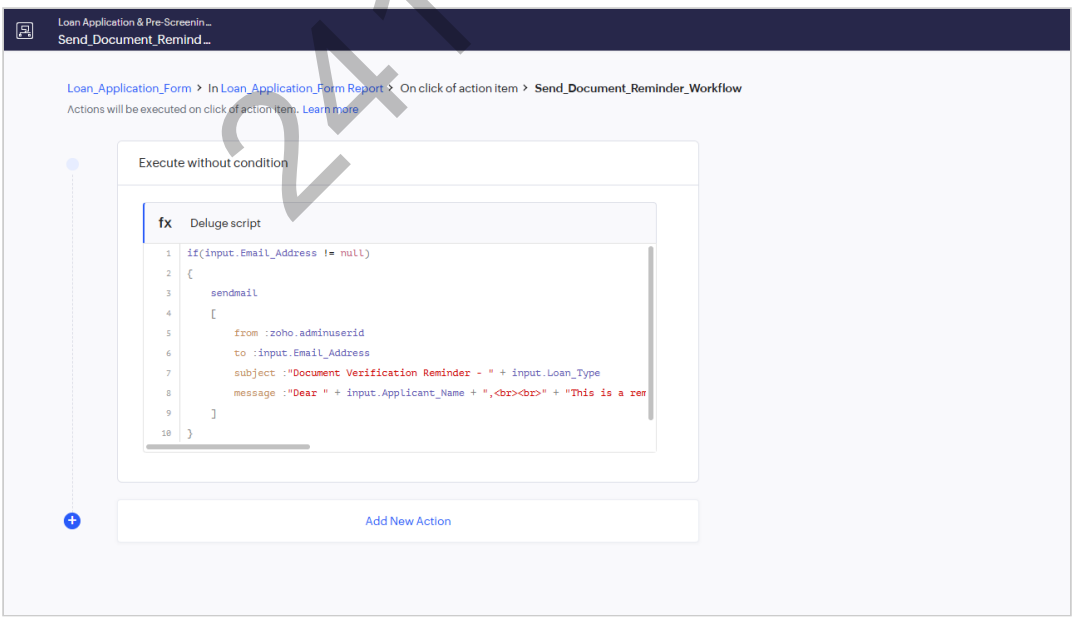


Figure 12: Script 3 in Send_Document_Reminder_Workflow (on click of action item)

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```
if(input.Email_Address != null)
{
    sendmail
    [
        from: zoho.adminuserid
        to: input.Email_Address
        subject: "Document Verification Reminder - " + input.Loan_Type
        message: "Dear " + input.Applicant_Name + ",<br><br>" + "This is a reminder
to please submit any pending documents for your " + input.Loan_Type + " application
so we can proceed with verification.<br><br>Regards,<br>Loan Application Team"
    ]
}
```

Script 3

3. Testing and Debugging

3.1 Test Cases

#	Test	Input	Expected	Actual	Result
1	Validation blocks an out-of-hours callback	Preferred_Callback_Time = 23:17:13	Alert shown; submission cancelled	Alert shown with the business-hours message; record not submitted (Figure 13)	Pass
2	Valid submission	Callback time within 09:00 to 17:59, all mandatory fields filled	Record saved; acknowledgement email sent	Record saved in the report; email received (Figure 14)	Pass
3	Reminder action	"Send Document Reminder" on a record with an email address	Reminder email sent to the applicant	Email sent; the action runs silently with no on-screen message	Pass

3.2 Debugging Log

#	Script	Code tried	Error	Outcome
1	Script 2	input.Applicant_Name.getPrefix() and .getSuffix()	"The number of arguments for 'getSuffix' function does not match the number of mandatory params '1'" (Figure 9)	Removed both calls; input.Applicant_Name is used directly and returns the full name.

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#	Script	Code tried	Error	Outcome
2	Script 1	Preferred_Callback_Time.toTime("HH").toLong()	"Value is empty and 'toLong' function cannot be applied", line 3 (Figure 6)	The conversion returned an empty value; replaced.
3	Script 1	Splitting the value on ":" with split()	Deluge reported that the split function could not be found (line 4)	Replaced.
4	Script 1	Preferred_Callback_Time.toString("HH").toLong()	Same "Value is empty" error, line 4 (Figure 7)	Replaced.
5	Script 1	Preferred_Callback_Time.subString(0,2).toLong()	None	Working. The form shows the Time value as HH:mm:ss text (for example 23:17:13), so its first two characters give the hour.

An unexpected result during testing: the acknowledgement email did not appear in the inbox at first. Because it is sent by an after-save workflow, a delivery problem produces no on-screen error, so the spam folder was checked and the email was found there.

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4. Output

The following figures show the final working behaviour of the automation.

Validation: submitting the form with a callback time of 23:17:13 is blocked and the business-hours message is displayed.

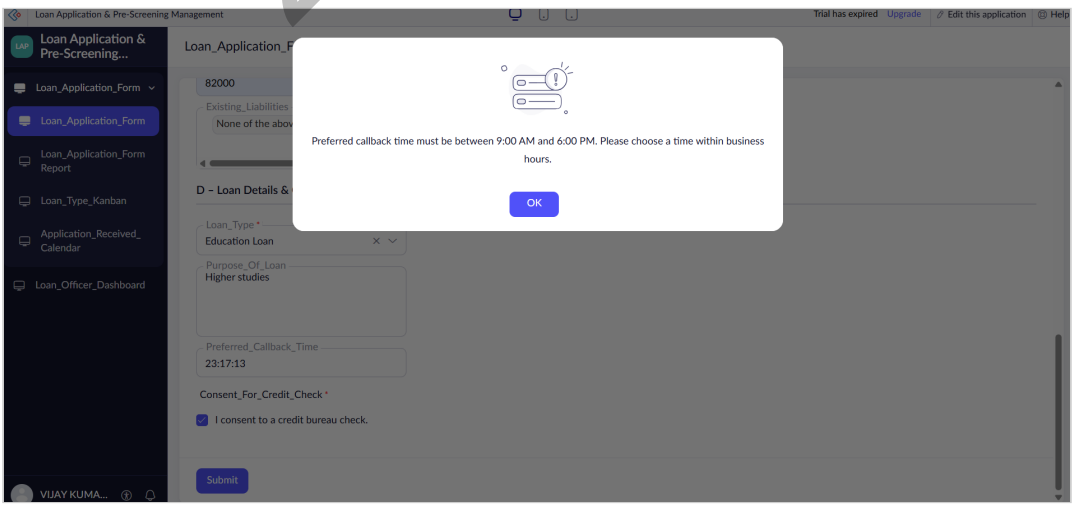


Figure 13: Validation alert blocking a submission with an out-of-hours callback time

Acknowledgement email: after a valid submission, the applicant receives the automated email with the loan type in the subject line and the applicant's name in the greeting. It was first delivered to the spam folder.

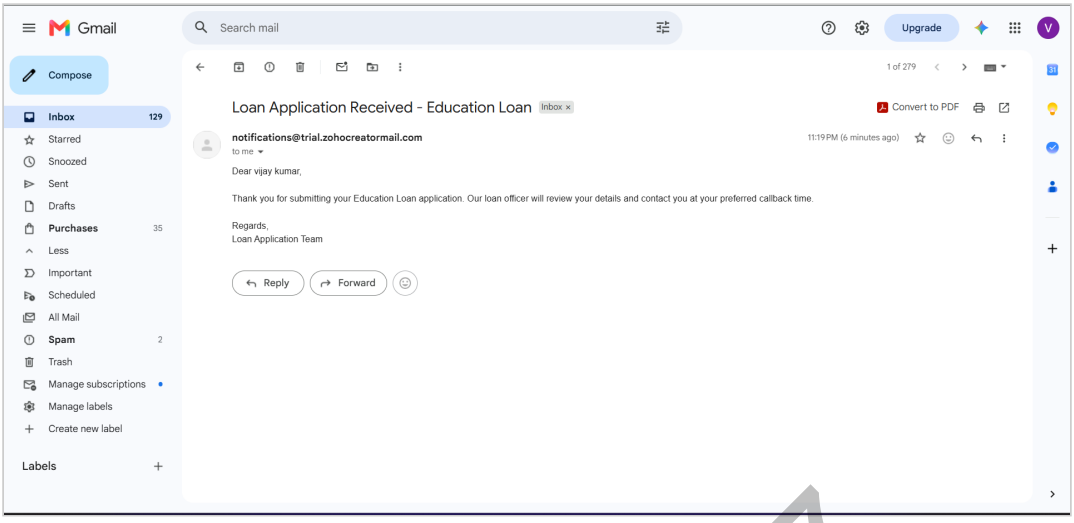


Figure 14: Acknowledgement email received after a successful submission

Officer reminder action: each record in the Loan_Application_Form Report now offers "Send Document Reminder" in its action menu. Selecting it runs Send_Document_Reminder_Workflow and emails the applicant.

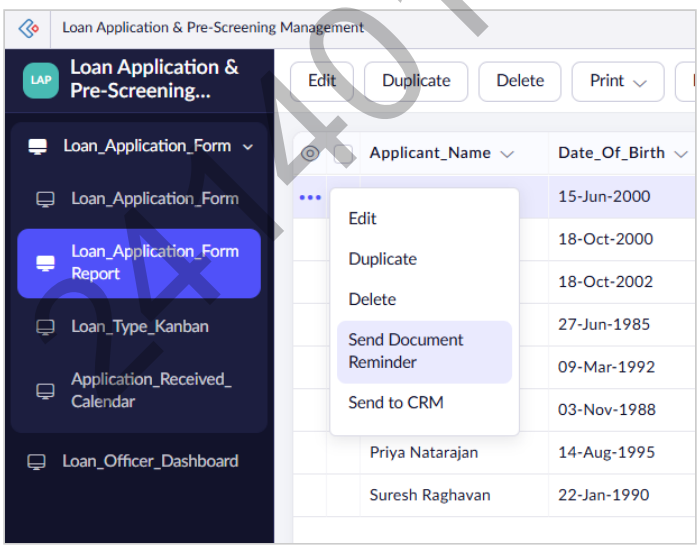


Figure 15: "Send Document Reminder" action in the Loan_Application_Form Report record menu

5. Result

The Loan Application & Pre-Screening Management application was extended with three Deluge-based workflows: a submission-time validation that rejects callback times outside business hours, an automatic acknowledgement email after a successful submission, and a report action that lets loan officers send a document-verification reminder. Testing exposed four script errors, which were debugged and fixed, and each workflow was then verified working.